Chapter 4 Intro to Networking Exam

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Score: 98% Passing Score: 80%

Question 1: ✔ Correct

You've connected a cable certifier to an RJ45 wall jack and the output shown below is displayed on the device. What does this output indicate? (Select two.)

```
1 ----------------------- 1
2 ----------------------- 2
3 ----------------------- 3
4 4
5 ----------------------- 5
6 6
7 ----------------------- 7
8 ----------------------- 8
```

- [ ] Pin 4 is shorted.
- [ ] Pin 6 is shorted.
- [ ] Pins 4 and 6 are straight-through.
- [ ] The cable is functioning correctly.
- [x] Pin 6 is open.
- [ ] Pins 4 and 6 are crossed-over.
- [x] Pin 4 is open.

EXPLANATION

In this example, any connections displayed with no characters between the pin numbers are open connections. This problem is usually caused by:

- Poor connections between the wire and the RJ45 jack.
- Individual wires broken within the UTP cable.

Output with x characters between pins indicates that they are shorted. Correctly functioning connections are displayed using - characters in the output of the cable certifier. Cross-over connections are displayed with lines between the crossed-over pins.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 2: ✔ Correct

Which Gigabit Ethernet standard uses multimode fiber optic cabling and supports network segments up to a maximum of 550 meters long?

- 1000BaseCX
- 1000BaseZX
- ➡ 1000BaseSX
- 1000BaseT

EXPLANATION

The 1000BaseSX standard uses multimode fiber optic cable with a maximum segment length of 550 meters. However, to implement segments this long, you must use 50-micron 500MHz/km multimode fiber optic cable. Other types of cable will shorten the maximum segment length. 1000BaseFX also supports lengths up to 550 meters using multimode cable. 1000BaseFX supports distances up to 10 kilometers using single mode cable.

1000BaseZX has a maximum segment length of up to 100 km. 1000BaseCX and 1000BaseT use copper cabling instead of fiber optic.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 3: ✔ Correct

You have been tasked with designing an Ethernet network. Your client needs to implement a very high-speed network backbone between campus buildings, some of which are around 300 meters apart. Multi-mode fiber optic cabling has already been installed between buildings. Your client has asked that you use the existing cabling.

Which Ethernet standard meets these guidelines? (Choose two.)

- [ ] 10BaseFL
- [x] 10GBaseSR
- [ ] 1000BaseT
- [x] 1000BaseSX
- [ ] 10GBaseER
- [ ] 1000BaseCX

EXPLANATION

10GBaseSR and 1000BaseSX can operate within these parameters. Both will support segment lengths 300 meters long and can use multi-mode fiber optic cabling.

10BaseFL isn't a good choice because its data transmission rate is relatively slow. 1000BaseCX and 1000BaseT both use copper wiring.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 4: ✔ Correct

You are moving a client to a new location within an Ethernet network. Previous to the move, the client system did not have difficulty accessing the network.

During the relocation, you attach patch cables from the client system to the wall jack and from the patch panel to the switch. Once connected, you do not get a link light on the network card or the switch. You swap out the cable running between the patch panel and the switch with a known working one, but you still cannot connect.

Which of the following might you suspect is the problem?

- Failed network card
- Failed switch
- Faulty termination
- Incorrect duplex settings
- Server software configuration error
- ➡ Failed patch cable between the client system and the wall jack

EXPLANATION

Because the client system previously worked in a different location, the issue is not likely related to the client system. The NIC and switch LEDs' failure to light indicate that there isn't an end-to-end connection between the client and the switch. This means that either the patch cable between the wall jack and the client is faulty or the cable between the patch panel and the switch faulty. The cable connecting the switch and the patch panel was verified, leaving the cable run between the wall jack and the client system.

It is unlikely that the issue is a failed network card, as the system was functioning in a different location, eliminating the possibility of hardware and software on the client system being the problem. The failure is limited to a connection issue between a single client and a switch. If the switch failed, all clients connected to that switch would not be able to connect. Incorrect duplex settings would not prevent NIC and switch LEDs from lighting.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 5: ✔ Correct

You want to connect the LAN port on a router to the uplink port on a switch. The switch does not support auto-MDI.

Which type of cable should you use?

- Loopback
- ➡ Crossover
- Straight-through
- Rollover

EXPLANATION

Use a crossover cable to connect a workstation or a router to the uplink port on a switch.

Use a straight-through cable to connect the router to a regular switch port. Use a rollover cable to connect a workstation to the console port of a router. Use a loopback plug to allow a device to communicate with itself through its own network adapter.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 6: ✔ Correct

What is the minimum cable specification that supports 1000 Mbps Ethernet?

- Cat 3
- Cat 4
- Cat 5
- ➡ Cat 5e
- Cat 6
- Cat 7

EXPLANATION

1000 Mbps Ethernet (Gigabit Ethernet) requires at least Cat 5e cables.

Cat 3 and Cat 4 only support 10 Mbps Ethernet. Cat 5 cable only supports up to 100 Mbps. Cat 6 or Cat 7 is required for bandwidth up to 10 Gbps Ethernet.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 7: ✔ Correct

You have a network connected using a physical star topology. One of the drop cables connecting a workstation has been removed.

Which of the following best describes what affect this will have on network communications?

- Devices on one side of the missing cable will be able to communicate with each other; devices on the other side of the missing cable will not be able to communicate.
- ➡ All devices will be able to communicate, except the device connected with the drop cable .
- Devices on one side of the missing cable will be able to communicate with each other; devices on the other side of the missing cable will be able to communicate with each other.
- All devices will be able to communicate.
- No devices will be able to communicate.

EXPLANATION

A break in a cable in a star means that the device connected to the central device (hub or switch) through that cable can no longer communicate on the network. All other hosts will be able to communicate with all other devices.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 8: ✔ Correct

This question includes an image to help you answer the question. View Image

You manage the two-location network shown in the exhibit. Workstations and servers at each location connect to a patch panel using behind-the-wall wiring. The patch panel then connects network hosts to one of three 1000BASE-T switches. Routers are implemented at each location to connect the two networks together using a private WAN. The switch ports have auto-MDIX disabled.

Drag the cable type on the left to the most appropriate network location on the right. Each cable type can be used more than once.

A. Drop cables

✔ Cat 6 straight-through UTP

B. Patch cables

✔ Cat 6 straight-through UTP

C. Uplink cables

✔ Cat 6 crossover UTP

D. WAN cables

✔ Single-mode fiber optic

EXPLANATION

In this scenario, the following cables should be used in the following locations:

- Drop cables = Cat 6 straight-through UTP.
- Patch cables = Cat 6 straight-through UTP.
- Uplink cables = Cat 6 crossover UTP (Cat 6 straight-through UTP can be used instead if auto-MDIX is enabled on the switch ports).
- WAN cable = Single-mode fiber optic.

Because 1000BASE-T Ethernet switches are in use at both locations, Cat 5 or better UTP must be used. Because the distance between locations is greater than 550 meters, single-mode fiber optic cable must be used. This will require the use of 1000BASE-LX or 1000BASE-EX transceivers on both routers.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 9: ✔ Correct

Which of the following use the CSMA/CD access method?

- 10BaseFL
- FDDI
- 1000BaseT
- Token Ring

EXPLANATION

CSMA/CD stands for carrier sense multiple access/collision detection. It defines the steps network devices take when two devices attempt to use a data channel simultaneously. Ethernet networks use CSMA/CD, including 1000BaseT.

REFERENCES

LabSim for Network Pro, Section 4.1.

Question 10: ✔ Correct

Users are complaining that sometimes network communications are slow. You use a protocol analyzer and find that packets are being corrupted as they pass through a switch. You also notice that this only seems to happen when the elevator is running.

What should you do?

- ➡ ◉ Install shielded cables near the elevator.
- ○ Add a UPS system to the switch.
- ○ Add a dedicated A/C unit to the switch closet.
- ○ Install a dedicated power circuit for the switch.

EXPLANATION

Interference is a signal that corrupts or destroys regular networking signals. Interference affects the availability of a network because normal communications are not possible. Sources of interference include elevators, generators, motors, and fluorescent lights.

Use a UPS or a dedicated power circuit to ensure that devices have constant power. Use a dedicated A/C unit to keep a server room or closet cool.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 11: ✔ Correct

You need to transfer data from one laptop to another, and you would like to use an Ethernet cable. You do not have a hub or a switch.

Which type of cable should you use?

- ➡ Crossover
- Loopback
- Rollover
- Straight-through

EXPLANATION

Use a crossover cable to connect two devices together in a back-to-back configuration.

Use a straight-through cable to connect a workstation to a hub or switch port. Use a rollover cable to connect a workstation to the console port of a router or a switch. Use a loopback plug to allow a device to communicate with itself through its own network adapter.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 12: ✔ Correct

This question includes an image to help you answer the question. View Image

Use the exhibit to match the connector type on the left with the corresponding letter on the right.

A

✔ DB25

B

✔ DB9

C

✔ RJ45

D

✔ RJ11

E

✔ LC

F

✔ BNC

G

✔ F-Type

EXPLANATION

- **A** DB25 connectors are older serial connectors.
- **B** DB9 connectors are usually on the ends of RS232 serial cables.
- **C** RJ45 connectors are used for Ethernet networking with twisted pair cables.
- **D** RJ11 connectors are used for dial-up and some DSL internet connections.
- **E** LC connectors are used with fiber optic cables.
- **F** BNC connectors are used with coaxial cables on 10Base2 Ethernet networks.
- **G** F-Type connectors are used with coaxial cables that create cable TV and broadband cable connections.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 13: ✔ Correct

Which of the following standards is typically used in a rollover cable?

- ➡ RS232
- RJ11
- RG58
- RG6

EXPLANATION

A rollover cable has a serial connector on one end and an RJ45 connector on the other end. RS232 is the standard for serial communications.

RJ11 connectors are used for analog telephone lines. RG6 and RG58 are coaxial cable standards.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 14: ✔ Correct

A network is connected following the IEEE 802.3 specifications. Which of the following best describes when a device can transmit messages?

- The device transmits whenever it is ready.
- The device is notified of its turn to send messages.
- The device requests permission from a controlling device.
- The device with the token can use the transmission media.
- ➡ The device listens to determine if the transmission media is free.

EXPLANATION

The IEEE 802.3 committee describes the CSMA/CD media access method. Devices listen to the network to determine if the transmission media is free before transmitting.

REFERENCES

LabSim for Network Pro, Section 4.1.

Question 15: ✔ Correct

During a network infrastructure upgrade, you have replaced two 10 Mbps hubs with switches and upgraded from Category 3 UTP cable to Category 5e. During the process, you accidentally cut the Cat 5e patch cable that stretches from the network printer to the upgraded switch.

What is the impact on the network?

- All network nodes connected to the switch will be unavailable.
- All network nodes authenticated by the same server as the printer will be unavailable.
- All network nodes on the same subnet as the printer will be unavailable.
- ➡ All network nodes except the printer will be available.
- All network nodes, including the printer, will be available.

EXPLANATION

UTP cable and switches are associated with a star network topology. In a star topology, each device is attached to the network using its own patch cable. If the cable were to fail for any reason, only the device connected by that cable would be unavailable.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 16: ✔ Correct

You have purchased a new router that you need to configure. You need to connect a workstation to the router's console port to complete the configuration tasks.

Which type of cable would you most likely use?

- Crossover
- Straight-through
- ➡ Rollover
- RG6

EXPLANATION

Use a rollover cable to connect a workstation to the console port of a router or a switch. The rollover cable has an RJ45 connector on one end to connect to the console port. The other end has a serial connector to connect to the serial port of the workstation. You then run a terminal emulation program on the workstation to connect to the console of the router or switch to perform configuration and management tasks.

Use a straight-through or crossover Ethernet cable to connect devices using the Ethernet RJ45 ports. An RG6 cable is a coaxial cable.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 17: ✔ Correct

Ethernet 100BaseFX networks use what type of cabling?

- Coaxial
- Shielded twisted pair
- ➡ Fiber optic
- Unshielded twisted pair

EXPLANATION

Ethernet 100BaseFX networks use fiber optic cabling.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 18: ✔ Correct

You have been tasked with designing a high-speed Ethernet network. Your client's building already has 150-ohm shielded twisted pair (STP) wiring installed. Due to budget constraints, they have asked you to reuse the existing wiring instead of installing new fiber optic cabling.

Which Ethernet standard could you implement in this situation?

- 1000BaseLX
- 1000BaseSX
- 1000BaseT
- 1000BaseCX
- 1000BaseZX
- 10BaseFL

EXPLANATION

The 1000BaseCX standard specifies 150-ohm STP cabling. The maximum cable length is 25 meters.

The 10BaseFL, 1000BaseSX, 1000BaseLX, and 1000BaseZX standards employ fiber optic cabling. 1000BaseT uses Category 5 UTP instead of STP cabling.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 19: ✔ Correct

Upon conducting a visual inspection of the server room, you see that a switch displays LED collision lights that are continually lit. You check the LED on the corresponding workstation and see that it is flashing rapidly even though it is not sending or receiving network traffic at that time.

What is the cause of the network collisions?

- Faulty switch
- ➡ Faulty network card
- PCI slot failure
- Adapter controller card failure
- Incorrect duplex settings

EXPLANATION

Sometimes when a NIC fails, it doesn't just stop working, but begins to flood the network with transmissions. This is called jabbering. A single network card can slow down and entire network by continually transmitting onto the network. A jabbering network card can be identified by a slower than normal network, by high occurrences of collisions displayed on the hub or switch, and by LEDs on a network card indicating a high level of transmissions even though a user is not using the network.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 20: ✔ Correct

You are planning a network for an educational campus. Due to the size of the buildings and the distance between them, you have elected to use 10BaseFL hubs, cabling, and network interface cards.

What is the maximum length for the network cable between a workstation and a hub?

- 220 meters
- 550 meters
- 1000 meters
- 412 meters
- 100 meters
- ➡ 2000 meters

EXPLANATION

The maximum length for a 10BaseFL network segment is 2000 meters (2 km). Because a 10BaseFL network uses a physical star topology, a segment is defined as one of the arms of the star (between the hub and a host). That means the fiber optic cable between the hub and a workstation can be up to 2000 meters long.

1000BaseSX and 1000BaseLX support multimode cable up to 550 meters. 100 meters is the maximum twisted pair cable length.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 21: ✔ Correct

You are creating an Ethernet network for your company. The shipping department is located in a different building that is located 150 meters from the main wiring closet. You connect a single Cat 6a cable to connect the wiring closet to the shipping building.

Which of the following should you include in your plan?

- Spanning tree
- Shielded twisted pair
- ➡ Repeater
- Half-duplex

EXPLANATION

A repeater regenerates the signal and removes the unwanted effects caused by attenuation. Attenuation is the loss of signal strength from one end of a cable to the other. In this example, the distance from the wiring closet to the other building exceeds the 100-meter maximum for Ethernet, so a repeater is necessary to regenerate the signal.

Use shielded twisted pair to protect against electromagnetic interference (EMI). Shielded cables have a metal foil that encloses all of the wires. Some cables might also include a drain wire, which is a bare wire in the middle of the wire bundle that helps to reduce EMI. More expensive cable might also use a metal foil around each pair of wires.

The spanning tree protocol runs on switches to prevent switching loops by making only a single path between switches active at a single time. When using half-duplex communications, devices perform collision detection because the transmission medium is shared.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 22: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 23: ✔ Correct

You want to create a rollover cable that has an RJ45 connector on both ends. How should you connect the wires within the connectors?

- Connect each pin on one end to the same pin on the other end (pin 1 to pin 1, pin 2 to pin 2, etc.).
- ➡ Connect pin 1 to pin 8, pin 2 to pin 7, pin 3 to pin 6, and pin 4 to pin 5.
- Connect pin 1 to pin 3 and pin 2 to pin 6.
- Use the T568A standard on one end and the T568B standard on the other end.

EXPLANATION

When terminated with an RJ45 connector on both ends, the wires within the connectors are rolled over to the opposite connector as follows:

- Pin 1 connects to pin 8
- Pin 2 connects to pin 7
- Pin 3 connects to pin 6
- Pin 4 connects to pin 5

A crossover cable uses the T568A standard on one end and the T568B standard on the other end. The crossover cable connects pin 1 to pin 3 and pin 2 to pin 6. Connecting each pin to the same pin on the other end creates a straight-through cable.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 24: ✔ Correct

Which Gigabit Ethernet standard can support long network segments up to a maximum of 5 km when used with single-mode fiber optic cable?

- 1000BaseSX
- ➡ 1000BaseLX
- 1000BaseT
- 1000BaseCX

EXPLANATION

1000BaseLX supports segment lengths of up to 5 km when used with single-mode fiber optic cable. This maximum segment length is cut to 550 m when multimode fiber optic cable is used.

1000BaseSX supports segment lengths of only 550 meters. 1000BaseCX uses copper wire and supports segment lengths of only 25 meters. 1000BaseT uses twisted pair cables.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 25: ✔ Correct

Which of the following physical topologies are used with Ethernet networks? (Select two.)

- [ ] Mesh
- ➡ [x] Star
- [ ] Ring
- ➡ [x] Bus

EXPLANATION

Ethernet networks use either a physical bus or physical star topology. Hubs can also be cascaded to form a tree topology.

REFERENCES

LabSim for Network Pro, Section 4.1.

Question 26: ✔ Correct

Angela is the network administrator for a rapidly growing company with a 100BaseT network. Users have recently complained about slow file transfers. While checking network traffic, Angela discovers a high number of collisions.

Which connectivity device would best reduce the number of collisions and allow future growth?

- Hub
- ➡ Switch
- Gateway
- Router
- Bridge

EXPLANATION

A switch would be the best choice in this situation. A bridge will segment traffic and reduce collisions, but it would be harder to maintain and harder to add new bridges as the network grows. A router would allow growth and reduce collisions. Switches can provide those benefits at a lower cost per port and offer more administration options.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 27: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 28: ✔ Correct

Which of the following connectors is typically used on one end of a rollover cable?

- LC
- SC
- Serial
- BNC
- F-type
- ST

EXPLANATION

A rollover cable has a serial connector on one end and an RJ45 connector on the other end. Alternatively, it might have an RJ45 connector on both ends and use a serial converter to convert from the RJ45 connector to a serial connector.

BNC and F-type connectors are used with coaxial cables. ST, SC, and LC connectors are used with fiber optic cables.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 29: ✔ Correct

Your network follows the 100BaseFX specifications for Fast Ethernet and uses half-duplex multi-mode cable. What is the maximum cable segment length allowed?

- 100 meters
- ➡ 412 meters
- 550 meters
- 1,000 meters
- 2,000 meters

EXPLANATION

100BaseFX half-duplex multimode cable has a maximum segment length of 412 meters.

1000BaseSX and 1000BaseLX support multimode cable up to 550 meters. 10BaseFL supports fiber optic cable between 1,000 and 2,000 meters.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 30: ✔ Correct

You have just connected four new computer systems to an Ethernet switch using spare patch cables. After the installation, only three systems are able to access the network. You verify all client network settings and replace the network card in the failed system. The client is still unable to access the network.

Which of the following might you suspect is the real cause of the problem?

- Incorrect LAN protocol
- Incorrect routing table
- ➡ Failed patch cable
- Failed switch
- Faulty IP stack

EXPLANATION

Bent and damaged patch cables will prevent client systems from accessing the network. In this scenario, a faulty patch cable is the most likely cause of the connection failure. The easiest way to test this is to simply swap out the cable and try a known working UTP patch cable.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 31: ✔ Correct

You are implementing a SOHO network for a local business. The ISP has already installed and connected a cable modem in the business.

The business has four computers that need to communicate with each other and the internet. The ISP's cable modem has only one RJ45 port. You need to set up the network within the following parameters:

- You must spend as little money as possible.
- You must not purchase unnecessary equipment.
- Computers need to have a gigabit connection to the network.
- New devices should not require management or configuration.

You examine each computer and notice only one of the four computers has a wireless NIC; they all have Ethernet NICs.

What should you purchase?

- A wireless AP and three new wireless NICs.
- A new cable modem with a built-in switch and CAT 6a cabling.
- A hub and CAT 5e cabling.
- ➡ An unmanaged switch and CAT 5e cabling.
- A managed switch and CAT 6 cabling.

EXPLANATION

You should purchase an unmanaged switch and CAT 5e cabling. Switches offer guaranteed bandwidth to each switch port and full-duplex communication. Unmanaged switches are autonomous in their function, requiring no port management or configuration. CAT 5e cabling supports transfer speeds up to 1000 Mbps.

Purchasing a new cable modem with a built-in switch would be more expensive than an unmanaged switch. Additionally, CAT6a cabling is unnecessary for this type of network. Because all of the computers already have wired NICs, purchasing a wireless AP and three new wireless NICs would introduce new costs. The wireless AP would also require additional management and configuration. Hubs suffer from collisions, so only half-duplex communication is possible. This wouldn't support 1000 Mbps speeds. A small business with four computers doesn't need the additional features that a managed switch provides.

REFERENCES

LabSim for Network Pro, Section 4.3.

Question 32: ✔ Correct

You would like to implement 10 Gbps Ethernet over a distance of 1 kilometer or greater. Which of the following would be the minimum requirement for the implementation? (Select two.)

- [x] 10GBaseLR standards
- [ ] Multi-mode fiber
- [ ] 10GBaseER standards
- [ ] 10GBaseSR standards
- [x] Single-mode fiber

EXPLANATION

For 10 Gbps at distances up to 10 kilometers, use 10GBaseLR with single-mode fiber.

Multi-mode fiber is cheaper, but has a shorter maximum distance than single-mode fiber. 10GBaseSR uses multi-mode fiber at distances up to 300 meters. 10GBaseER supports distances up to 40 kilometers using single-mode fiber.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 33: ✔ Correct

Which of the following are requirements of the 1000BaseT Ethernet standards? (Select three.)

- [x] The cable length must be less than or equal to 100m
- [ ] Cat 5 cabling
- [x] RJ45 connectors
- [ ] The cable length must be less than or equal to 1000m
- [ ] SC or ST connectors
- [x] Cat 5e cabling

EXPLANATION

Gigabit Ethernet (1000BaseT) has similar requirements to 100BaseT with connectors, cabling, and distances. The network cards are simply designed to transfer data ten times as fast.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 34: ✔ Correct

A user from the sales department calls to report that he is experiencing problems connecting to the sales file server. All users in the sales department connect to the sales server through a single Ethernet switch. No other users have reported problems connecting to the sales server.

Which of the following troubleshooting actions are you most likely to perform first?

- Reinstall the network card drivers on the sales server.
- Replace the Ethernet switch in the sales department.
- ➡ Replace the network card in the user's computer.
- Replace the network card in the sales server.

EXPLANATION

In this scenario, you are most likely to replace the network card in the user's computer.

As there is only one user experiencing a problem, you are unlikely to replace the network card in the server or replace the Ethernet switch. For the same reason, you are also unlikely to replace the network card drivers on the server. If more than one user were experiencing the problem, any of the options could be a valid troubleshooting step.

REFERENCES

LabSim for Network Pro, Section 4.4.

Question 35: ✕ Incorrect

The media access control method of all Ethernet networks is _____.

- ~~CSMA/CA~~
- Token passing
- ➡ CSMA/CD
- Polling

EXPLANATION

Carrier sense multiple access with collision detection (CSMA/CD) is the media access control method of all Ethernet networks.

REFERENCES

LabSim for Network Pro, Section 4.1.

Question 36: ✔ Correct

What type of cabling is used with 100BaseTX Fast Ethernet networks?

- ➡ Cat5 or higher
- Fiber optic
- Cat3
- Cat4

EXPLANATION

Cat5 Ethernet cable or higher can be used with 100BaseTX Fast Ethernet networks.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 37: ✔ Correct

Your network follows the 100BaseTX specifications for Fast Ethernet. What is the maximum cable segment length allowed?

- ➡ 100 meters
- 412 meters
- 1,000 meters
- 2,000 meters
- 500 meters

EXPLANATION

Fast Ethernet using twisted pair cables (either 100BaseT4 or 100BaseTX) has a maximum cable segment length of 100 meters.

All Ethernet networks that use twisted pair cable (Ethernet, Fast Ethernet, Gigabit Ethernet) have a distance limitation of 100 meters.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 38: ✔ Correct

What topology is used with 100BaseTX Fast Ethernet networks? (Select two.)

- [ ] Physical ring/logical star
- [x] Physical star/logical star
- [ ] Physical star/logical ring
- [x] Physical star/logical bus

EXPLANATION

100BaseTX Fast Ethernet networks use a physical star/logical bus topology when a hub is used or a physical star/logical star when a switch is used.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 39: ✔ Correct

Which of the following Ethernet standards uses fiber-optic cabling? (Select two.)

- [ ] 1000BaseCX
- [ ] 100BaseT4
- [x] 100BaseFX
- [ ] 100BaseTX
- [x] 1000BaseLX

EXPLANATION

100BaseFX and 1000BaseLX are Ethernet standards that use fiber optic cabling. Following the Ethernet naming conventions:

- F designates fiber-optic cables. Ethernet standards with the F designation are 10BaseFL and 100BaseFX.
- L designates long distances and requires fiber-optic to support the distance. Ethernet standards with the L designation are 10BaseFL, 1000BaseLX, and 10GBaseLR.
- S designates short distances that use fiber-optic cables. Ethernet standards with the S designation are 1000BaseSX and 10GBaseSR.
- T designates twisted pair cables. Ethernet standards with the T designation are 10BaseT, 100BaseTX, 100BaseT4, and 1000BaseT.
- C designates copper cables. The 1000BaseCX standard is for fast Ethernet at short distances within wiring closets.

REFERENCES

LabSim for Network Pro, Section 4.2.

Question 40: ✔ Correct

You need to connect a workstation to a switch using a regular port on the switch (not an uplink port). The switch does not support auto-MDI.

Which type of cable should you use?

- Rollover
- ➡ Straight-through
- Crossover
- Loopback

EXPLANATION

Use a straight-through cable to connect a workstation or router to a regular switch port.

Use a crossover cable to connect the workstation to the uplink port. Use a rollover cable to connect the workstation to the console port of the switch. Use a loopback plug to allow a workstation to communicate with itself through its own network adapter.

REFERENCES

LabSim for Network Pro, Section 4.3.

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